

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA5112 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO5: *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO (BL5 , BL6)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A

PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

Code of Conduct:

I _E. Lakshmi Narayanan(192524249)_ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

E. Lakshmi Narayanan

1. Design pseudocode for a **stateful firewall system that tracks active connections and filters packets based on session state. Evaluate how state tracking improves security compared to stateless filtering.**

Pseudocode:

START

Create connection table

For each incoming packet

 Check packet details in connection table

 If connection is already active

 Allow packet

 Else if packet follows firewall rules

 Allow packet

 Add connection to table

 Else

 Block packet

 End If

End For

STOP

Evaluation:

Stateful firewalls are more secure than stateless firewalls because they remember active connections and can block unexpected packets.

2. Design pseudocode for a **signature-based malware detection system that scans files and compares them against a known signature database. Evaluate its effectiveness in detecting known threats and its limitations against new attacks.**

Pseudocode:

START

Load known malware signatures

Read the file

Generate file signature

If file signature matches a known signature

 Display "Malware Detected"

 Block the file

Else

 Display "File is Safe"

End If

STOP

Evaluation:

It is very effective for detecting known malware, but it may not detect new or modified malware whose signature is not present in the database.

3. Design pseudocode for a **distributed denial-of-service (DDoS) detection system using traffic threshold analysis. Evaluate how the algorithm differentiates between legitimate high traffic and attack traffic.**

Pseudocode:

START

Set traffic threshold

Monitor network traffic

If traffic exceeds the threshold

 Check source addresses and traffic pattern

 If traffic is abnormal

 Display "Possible DDoS Attack"

 Block or limit traffic

 Else

 Display "High but Legitimate Traffic"

 End If

Else

 Display "Normal Traffic"

End If

STOP

Evaluation:

Threshold analysis can quickly detect abnormal traffic. However, legitimate events such as a large number of users accessing a website may also produce high traffic and cause false alarms.

4. Design pseudocode for a network intrusion detection system using machine learning classification. Evaluate how feature selection and training data impact detection accuracy.

Pseudocode:

START

Collect network traffic data

Select important features

Split data into training and testing data

Train the machine learning model

For each new network packet

 Extract features

 Predict packet as Normal or Attack

 If prediction is Attack

 Generate alert

 Else

 Allow packet

 End If

End For

STOP

Evaluation:

Good feature selection and high-quality training data improve detection accuracy. Poor or incomplete training data can increase false detections.

5. Design pseudocode for a secure network access control system that authenticates users and enforces role-based access policies. Evaluate how access control mechanisms enhance overall network security.**Pseudocode:**

START

Enter username and password

If username and password are correct

 Identify user role

 If role has permission

 Grant access

 Else

 Deny access

 End If

Else

 Deny access

End If

STOP

Evaluation:

Role-based access control improves security by allowing users to access only the resources required for their roles. It reduces unauthorized access and protects sensitive network resources.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient